MEDISAARTHI

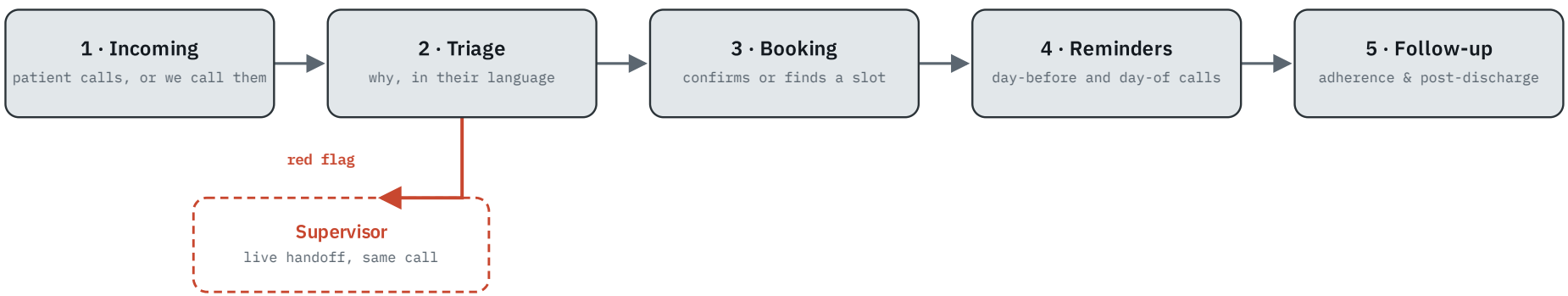
How it works

MediSaarthi is a phone-based AI assistant that talks to patients in their own language, handles the admin work hospitals struggle to staff for — triage questions, booking, reminders, follow-up calls — and hands off to a real person the instant anything sounds medically serious. It's an independent product — its own backend, frontend and database — built using the same call-handling patterns Novalantis's TalklyAI platform uses, but not its code.

Some of this already works today, in this repo. Some of it is still being built. This document walks through the whole thing once in plain terms, then again as an architecture diagram — and says clearly, throughout, which parts are which.

01 A patient's call, step by step

One conversation thread, from the first ring to the last follow-up call — with a human supervisor always reachable, not just notified afterward.



STEP 1

Incoming

The patient dials the hospital's number, or the hospital's schedule triggers an outbound call (a reminder, an adherence check). Either way, the system already knows which hospital and which language to use.

`built — Twilio → Bolna routing works today`

STEP 2

Triage

A short conversation about why the patient is calling — never a diagnosis. Every sentence is checked against that hospital's own red-flag symptom list as it's said. A match hands the call to a person immediately, mid-conversation.

`rule check built; live handoff still unconfirmed`

STEP 3

Booking

Confirms, reschedules, or books a slot off the hospital's live calendar — while the patient is still on the line, so a cancelled slot can be offered to the next person right away.

`built — atomic slot booking, not yet tested live`

STEP 4

Reminders

Day-before and day-of calls in the patient's language, with the ability to confirm, reschedule or cancel right there on the call instead of needing a second one.

`trigger endpoint built; scheduling not built`

STEP 5

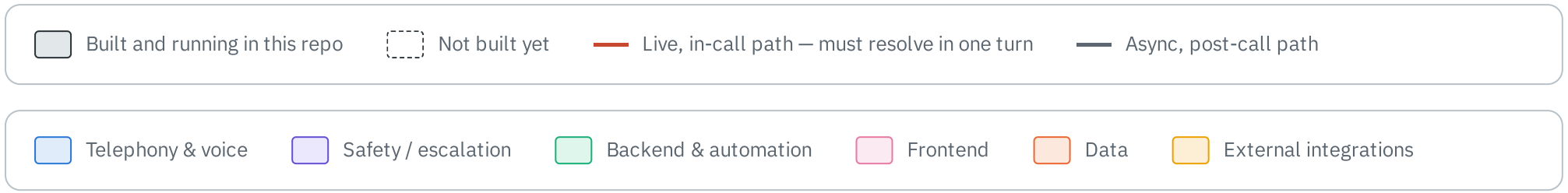
Follow-up

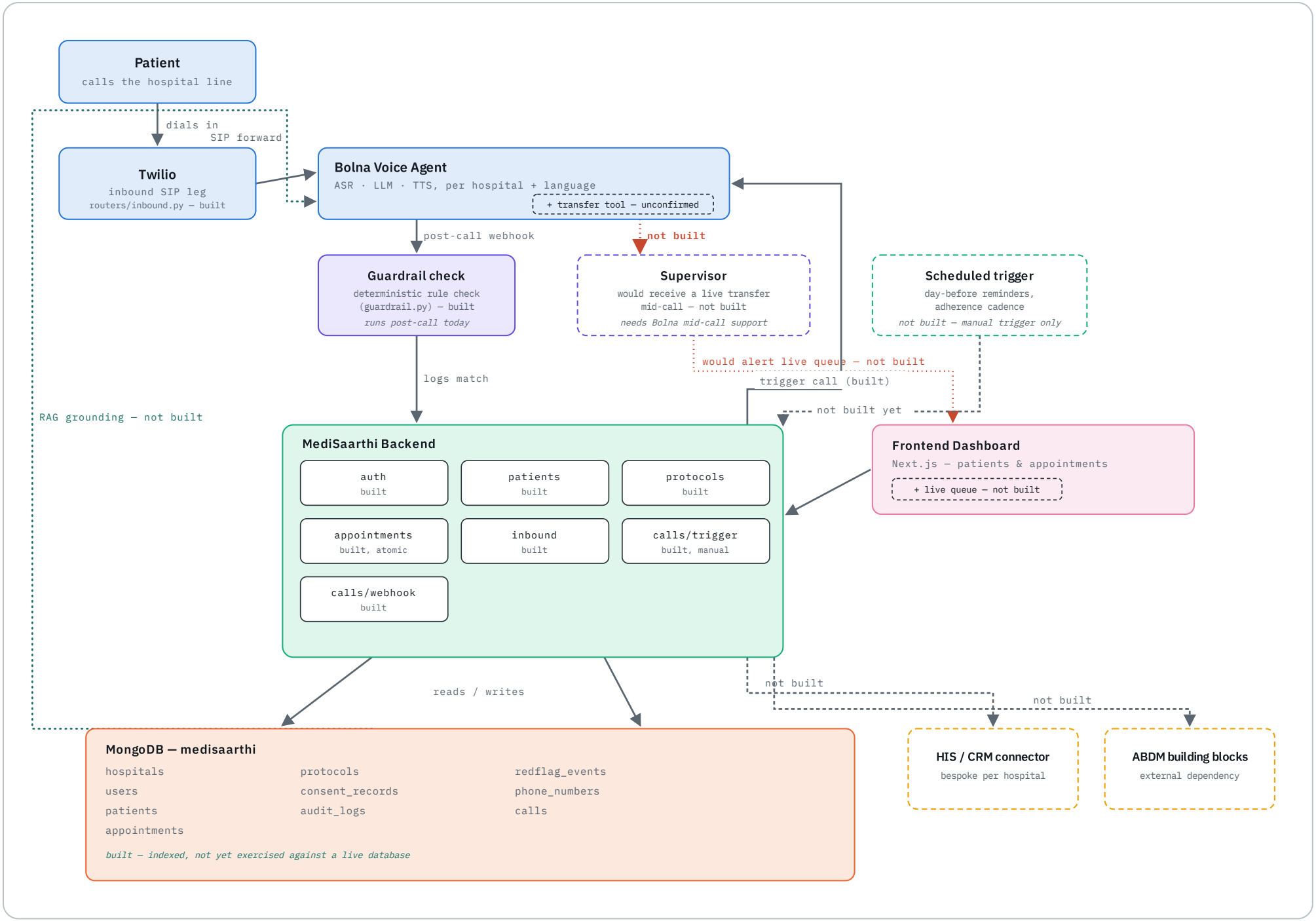
Scheduled medication reminders and post-discharge check-ins on a recurring cadence, so a missed response shows up for a supervisor instead of silently disappearing.

```
same trigger endpoint; no cadence or alerting yet
```

02 The full picture

Same flow, drawn as the system that runs it. Every box is either built and running in this repo today, or it isn't yet — the border style says which. The red path is the one that has to finish inside a single conversation turn; everything else can afford to be a webhook that runs after the call ends.




Patient
calls the hospital line
dials in
SIP forward
Twilio
inbound SIP leg
routers/inbound.py — built
Bolna Voice Agent
ASR · LLM · TTS, per hospital + language
+ transfer tool — unconfirmed
post-call webhook
not built
Guardrail check
deterministic rule check (guardrail.py) — built
runs post-call today
Supervisor
would receive a live transfer mid-call — not built
needs Bolna mid-call support
Scheduled trigger
day-before reminders, adherence cadence
not built — manual trigger only
would alert live queue — not built
trigger call (built)
not built yet
logs match
RAG grounding — not built
MediSaarthi Backend
auth
built
patients
built
protocols
built
appointments
built, atomic
inbound
built
calls/trigger
built, manual
calls/webhook
built
Frontend Dashboard
Next.js — patients & appointments
+ live queue — not built
reads / writes
not built
not built
MongoDB — medisaarthi
hospitals
users
patients
appointments
protocols
consent_records
audit_logs
redflag_events
phone_numbers
calls
built — indexed, not yet exercised against a live database
HIS / CRM connector
bespoke per hospital
ABDM building blocks
external dependency

Solid, filled boxes are built and running in this repo today. Dashed, outline boxes — and dashed red arrows — are not built yet. This is MediSaarthi's own independent backend; nothing here is imported from Talkly.